In [1]:
```
# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:
```
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.stattools import adfuller
```

In [3]:
```
# LOAD DATASET

df = pd.read_csv('AirPassengers.csv')
```

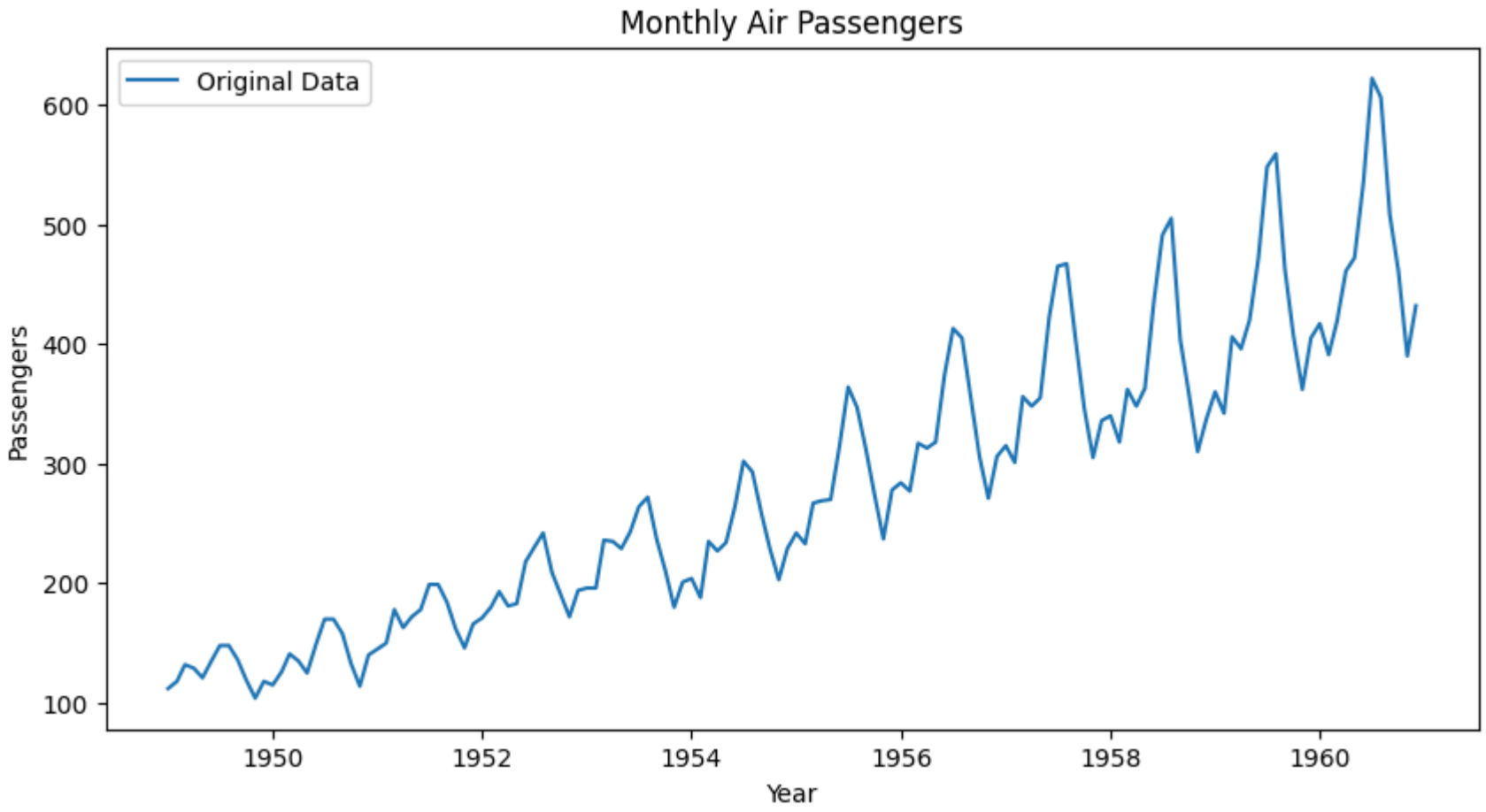
In [4]:
```
# Convert Month column to datetime
df['Month'] = pd.to_datetime(df['Month'])
df.set_index('Month', inplace=True)
```

In [5]:
```
print(df.head())
```

```
            #Passengers
Month
1949-01-01          112
1949-02-01          118
1949-03-01          132
1949-04-01          129
1949-05-01          121
```

In [6]:
```
# 2. VISUALIZE DATA

plt.figure(figsize=(10,5))
plt.plot(df, label='Original Data')
plt.title("Monthly Air Passengers")
plt.xlabel("Year")
plt.ylabel("Passengers")
plt.legend()
plt.show()
```



In [7]:
```
# 3. CHECK STATIONARITY (ADF TEST)

def adf_test(series):
    result = adfuller(series)
    print("\nADF Test Result:")
    print("ADF Statistic:", result[0])
    print("p-value:", result[1])
    if result[1] <= 0.05:
        print("Data is Stationary")
    else:
        print("Data is NOT Stationary")
```

```
adf_test(df['#Passengers'])
```

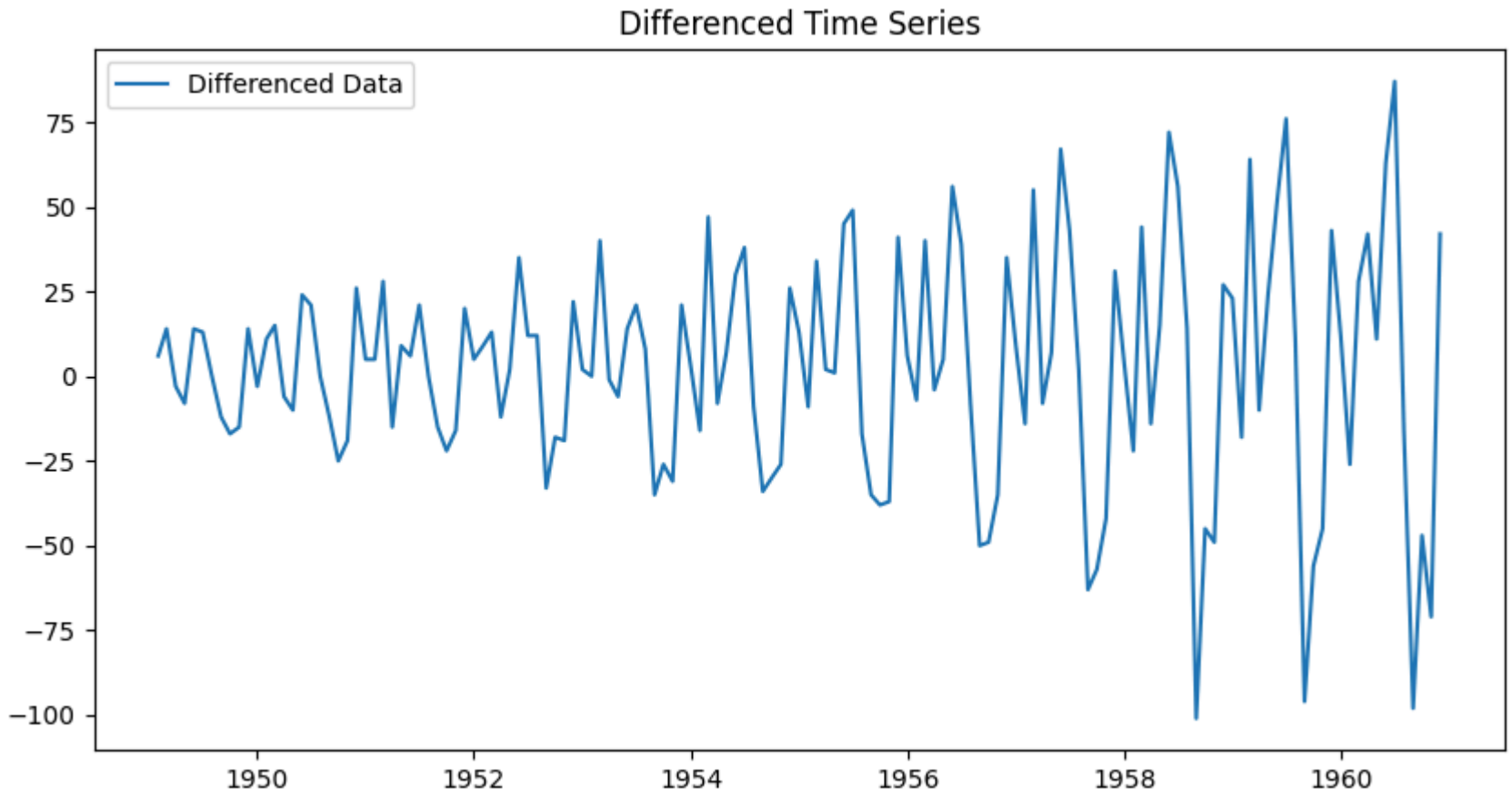
```
ADF Test Result:
ADF Statistic: 0.8153688792060482
p-value: 0.991880243437641
Data is NOT Stationary
```

In [8]:

```
# 4. MAKE DATA STATIONARY (DIFFERENCING)

df_diff = df.diff().dropna()

plt.figure(figsize=(10,5))
plt.plot(df_diff, label='Differenced Data')
plt.title("Differenced Time Series")
plt.legend()
plt.show()
```



In [9]:
```
# 5. TRAIN TEST SPLIT


train = df[:'1958']
test = df['1959':]

print("\nTrain size:", len(train))
print("Test size:", len(test))
```

```
Train size: 120
Test size: 24
```

In [10]:
```
# 6. BUILD ARIMA MODEL

# (p,d,q) = (5,1,0)

model = ARIMA(train['#Passengers'], order=(5,1,0))
model_fit = model.fit()

print(model_fit.summary())
```

```
                               SARIMAX Results
==============================================================================
Dep. Variable:            #Passengers   No. Observations:                  120
Model:                 ARIMA(5, 1, 0)   Log Likelihood                -554.265
Date:                Sat, 18 Apr 2026   AIC                           1120.531
Time:                        23:43:53   BIC                           1137.206
Sample:                    01-01-1949   HQIC                          1127.302
                         - 12-01-1958
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1          0.3015      0.097      3.101      0.002       0.111       0.492
ar.L2         -0.2301      0.106     -2.165      0.030      -0.438      -0.022
ar.L3         -0.0496      0.068     -0.734      0.463      -0.182       0.083
ar.L4         -0.3087      0.091     -3.384      0.001      -0.487      -0.130
ar.L5          0.0441      0.119      0.372      0.710      -0.188       0.277
sigma2       646.7213     92.082      7.023      0.000     466.244     827.198
===================================================================================
Ljung-Box (L1) (Q):                   0.00   Jarque-Bera (JB):                 0.23
Prob(Q):                              0.96   Prob(JB):                         0.89
Heteroskedasticity (H):               6.93   Skew:                             0.06
Prob(H) (two-sided):                  0.00   Kurtosis:                         3.18
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
```

```
d:\Applied Data Science Lab\.venv\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarning: No frequency informati
on was provided, so inferred frequency MS will be used.
  self._init_dates(dates, freq)
d:\Applied Data Science Lab\.venv\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarning: No frequency informati
on was provided, so inferred frequency MS will be used.
  self._init_dates(dates, freq)
d:\Applied Data Science Lab\.venv\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: ValueWarning: No frequency informati
on was provided, so inferred frequency MS will be used.
  self._init_dates(dates, freq)
```

In [11]:

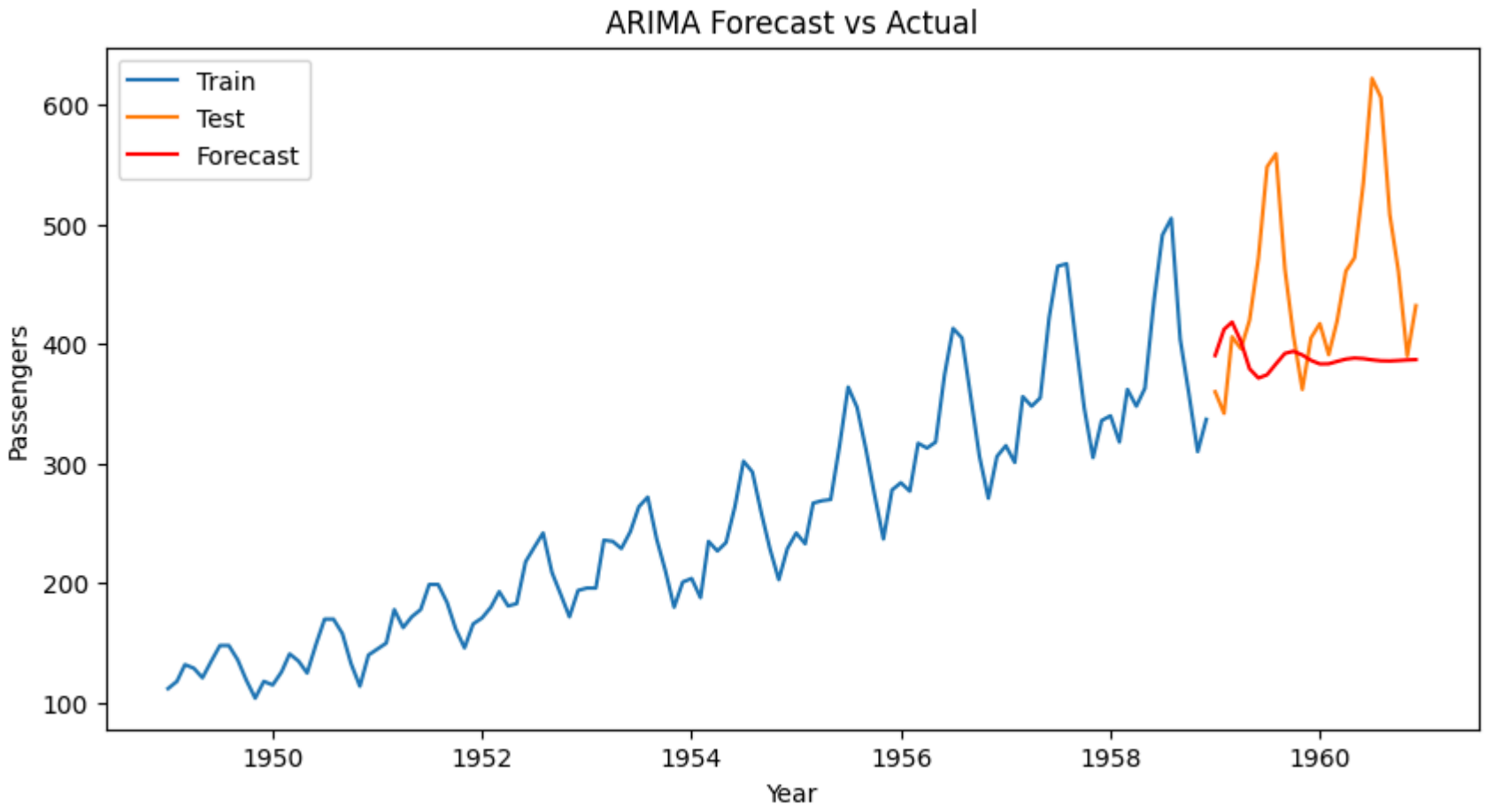
```
# 7. FORECAST

forecast = model_fit.forecast(steps=len(test))
```

In [12]:

```
# 8. PLOT RESULTS

plt.figure(figsize=(10,5))
plt.plot(train, label='Train')
plt.plot(test, label='Test')
plt.plot(test.index, forecast, label='Forecast', color='red')
plt.title("ARIMA Forecast vs Actual")
plt.xlabel("Year")
plt.ylabel("Passengers")
plt.legend()
plt.show()
```



In [13]:
```
# 9. EVALUATION

from sklearn.metrics import mean_squared_error

mse = mean_squared_error(test, forecast)
rmse = np.sqrt(mse)

print("\nModel Evaluation:")
print("RMSE:", rmse)
```

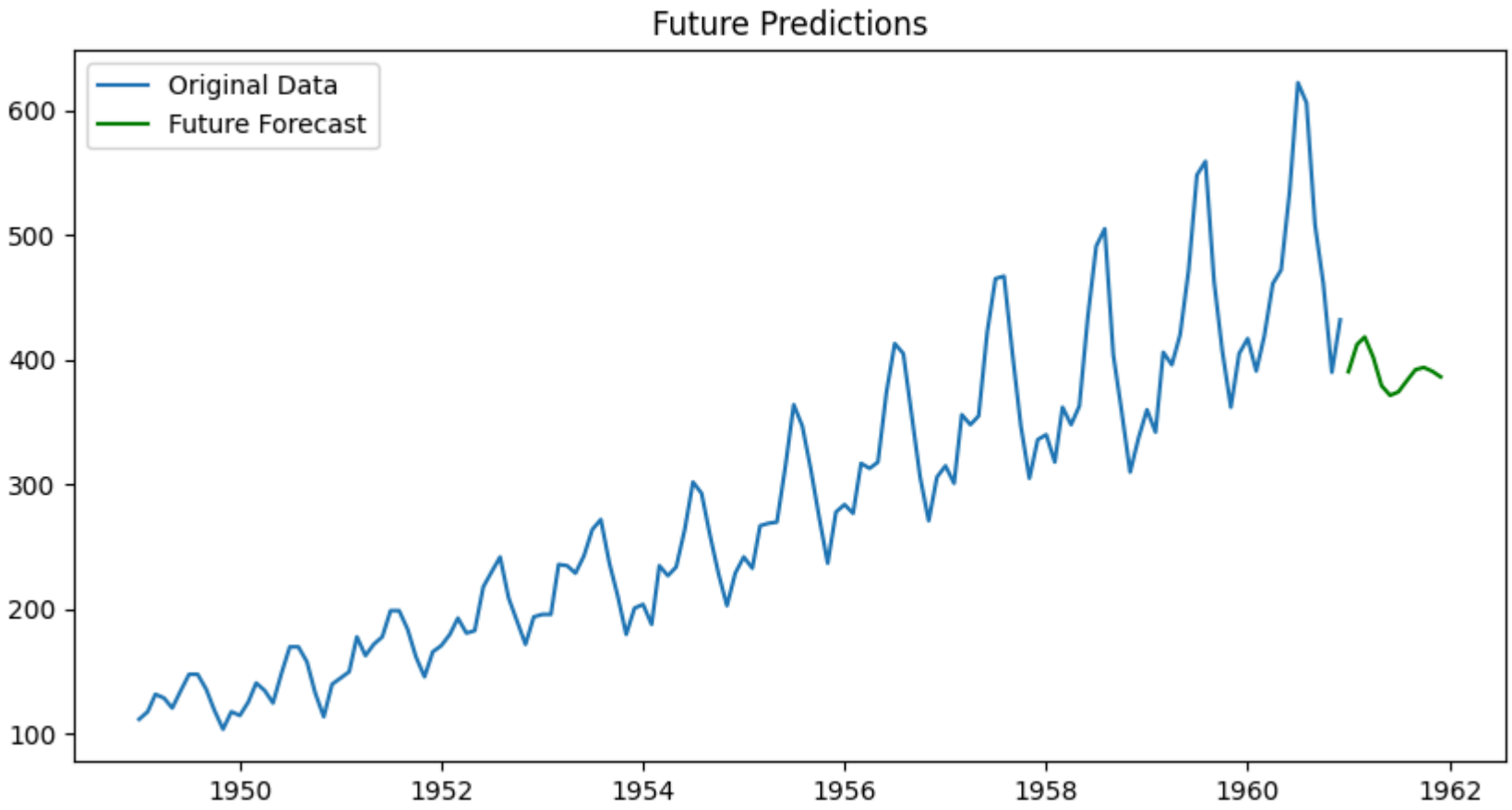
```
Model Evaluation:
RMSE: 101.65923228779866
```

In [14]:
```
# 10. FUTURE PREDICTION (NEXT 12 MONTHS)
# ================================
future_forecast = model_fit.forecast(steps=12)

print("\nNext 12 Months Forecast:")
print(future_forecast)

plt.figure(figsize=(10,5))
plt.plot(df, label='Original Data')
future_index = pd.date_range(start=df.index[-1], periods=12, freq='ME')
plt.plot(future_index, future_forecast, label='Future Forecast', color='green')
plt.legend()
plt.title("Future Predictions")
plt.show()
```

```
Next 12 Months Forecast:
1959-01-01    390.436616
1959-02-01    412.193788
1959-03-01    418.256657
1959-04-01    401.933514
1959-05-01    379.237324
1959-06-01    371.493515
1959-07-01    374.279405
1959-08-01    383.331769
1959-09-01    392.088407
1959-10-01    393.895717
1959-11-01    390.775333
1959-12-01    386.313674
Freq: MS, Name: predicted_mean, dtype: float64
```


Future Predictions
Original Data
Future Forecast
600
500
400
300
200
100
1950
1952
1954
1956
1958
1960
1962

In [ ]: